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Week 2 Module:

1. Find Student Roll Number (Linear Search)

Problem statement
Your class teacher has list of student roll number stored in the order they registered. Since the list is unsorted, you need to find whether a particular student roll number exists.

Write a program to search for the given roll number using linear search.

```
#include <stdio.h>

int main () {
    int n, x, i;
    scanf("%d", &n);

    int arr[n];

    for (i=0; i<n; i++) {
        scanf("%d", &arr[i]);
    }

    Scanf("%d", &x);

    for (i=0; i<n; i++) {
        if (arr[i]==x) {
            printf ("Found at index %d", i);
            return 0;
        }
    }

    printf("Not Found");
    return 0;
```

C linearsearch.c ×

Chapter Hello World > C linearsearch.c > main()

```c
#include <stdio.h>

int main() {
    int n, x, i;
    scanf("%d", &n);

    int arr[n];

    for(i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    scanf("%d", &x);

    for(i = 0; i < n; i++) {
        if(arr[i] == x) {
            printf("Found at index %d", i);
            return 0;
        }
    }

    printf("Not Found");
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
abhishekmishra28@Abhisheks-MacBook-Air Chapter Hello World % gcc linearsearch.c
abhishekmishra28@Abhisheks-MacBook-Air Chapter Hello World % ./a.out
5
101 105 113 110 130
110
Found at index 3%
abhishekmishra28@Abhisheks-MacBook-Air Chapter Hello World %
```

Output:-

5

101 105 103 110 1030

110

Found at index 3

7. Search Product ID (Binary Search)

Problem Statement

An online store keeps product IDs sorted in ascending order. A customer wants to know whether a particular products exists in inventory.

Write a program to search the product using Binary Search.

```
#include <stdio.h>

int main() {
    int n, target;
    scanf("%d", &n);

    int arr[n];

    for (int i=0; i<n; i++) {
        scanf("%d", &arr[i]);
    }

    scanf("%d", &target);

    int low = 0; high = n-1;

    while (low <= high) {
        int mid = (low+high)/2;
```

C binarysearch.c ×

Chapter Hello World > C binarysearch.c > main()

```c
#include <stdio.h>

int main() {
    int n, target;
    scanf("%d", &n);

    int arr[n];

    for(int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    scanf("%d", &target);

    int low = 0, high = n - 1;

    while(low <= high) {
        int mid = (low + high) / 2;

        if(arr[mid] == targe (char [18])"Product Available"
            printf("Product Available");
            return 0;
        }
        else if(arr[mid] < target)
            low = mid + 1;
        else
            high = mid - 1;
    }

    printf("Product Not Available");
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
abhishekmishra28@Abhisheks-MacBook-Air Chapter Hello World % gcc binarysearch.c
abhishekmishra28@Abhisheks-MacBook-Air Chapter Hello World % ./a.out
6
10 20 30 40 50 60
40
Product Available%
abhishekmishra28@Abhisheks-MacBook-Air Chapter Hello World %
```

```
if (arr[mid] = = target) {
    printf(" Product Available");
    return 0;
  }
  else if (arr[mid] < target)
      low = mid+1;
  else
      high = mid -1;
  }
  printf ("Product Not Available");
  return 0;
}
```

Output :-

```
6
10 20 30 40 50 60
40
Product Available
```

3. Arrange Exam Scores (Bubble sort)

Problem statement

A teacher wants to arrange students exam scores in ascending order to prepare the result sheet. Write a program using Bubble Sort.

Write a program using Bubble Sort.

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];

    for(int i=0; i<n; i++){
        scanf("%d", &arr[i]);
    }

    for(int i=0; i<n-1; i++){
        for( int j=0; j<n-i-1; j++){
            if (arr[j] > arr [j+1]){
                int temp = arr [j];
                arr [j] = arr[j+1];
                arr [j+1] = temp;
            }
        }
    }

    for (int i=0; i<n; i++){
        print("%d", arr [i]);
    }
    return 0;
}
```

Output

5

89 45 78 12 99

12 45 79 89 99

C bubblesort.c ×

Chapter Hello World > C bubblesort.c > main()

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];

    for(int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    for(int i = 0; i < n - 1; i++) {
        for(int j = 0; j < n - i - 1; j++) {
            if(arr[j] > arr[j + 1]) {
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }

    for(int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }

    return 0;
}
```

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

```
abhishekmishra28@Abhisheks-MacBook-Air Chapter Hello World % gcc bubblesort.c
abhishekmishra28@Abhisheks-MacBook-Air Chapter Hello World % ./a.out
5
89 45 78 12 99
12 45 78 89 99 %
abhishekmishra28@Abhisheks-MacBook-Air Chapter Hello World %
```

4. Arrange Library Book Numbers (Insertion Sort)

Problem Statement

A librarian receives book serial numbers one by one and wants to place each new book in the correct position.

Write a program using Insertion Sort to arrange book number

```
#include <stdio.h>

int main() {
        int n;
        scanf("%d", &n);

       int arr[n];

       for(int i=0; i<n; i++) {
           scanf("%d", &arr[i]);
       }

      for(int i=1; i<n; i++){
          int key = arr[i];
          int j = i-1;

          while(j>=0 && arr[j] >key){
              arr[j+1] = arr[j];
              j--;
          }

          arr[j+1] = key;
      }

      for(int i=0; i<n; i++){
          printf("%d", arr[i]);
      }
      return 0;
}
```

C insertionsort.c ×

Chapter Hello World > C insertionsort.c > main()

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];

    for(int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    for(int i = 1; i < n; i++) {
        int key = arr[i];
        int j = i - 1;

        while(j >= 0 && arr[j] > key) {
            arr[j + 1] = arr[j];
            j--;
        }

        arr[j + 1] = key;
    }

    for(int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }

    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
abhishekmishra28@Abhisheks-MacBook-Air Chapter Hello World % gcc insertionsort.c
abhishekmishra28@Abhisheks-MacBook-Air Chapter Hello World % ./a.out
5
55 23 87 11 34
11 23 34 55 87 %
abhishekmishra28@Abhisheks-MacBook-Air Chapter Hello World %
```

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Output:

5

55 23 87 11 34

11 23 34 55 87

5. Rank Players by score (selection sort)

Problem Statement

A gaming club wants to rank players based on scores from lowest to highest.

Write a program using selection Sort.

```
#include<stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];

    for (int i=0; i<n; i++) {
        scanf("%d", &arr[i]);
    }

    for (int i = 0; i<n-1; i++) {
        int minIndex = i;

        for (int j = i+1; j<n; j++) {
            if (arr[j] < arr[minIndex]) {
                minIndex = j;
            }
        }
```

```
int temp = arr[i];
arr[i] = arr[minIndex];
arr[minIndex] = temp;
}
for (int i=0; i<n; i++) {
    printf("%d", arr[i]);
}

return0;
}
```

Output:-

5

70 20 90 10 40

10 20 40 70 90

C selectionsort.c ×

Chapter Hello World > C selectionsort.c > main()

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];

    for(int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    for(int i = 0; i < n - 1; i++) {
        int minIndex = i;

        for(int j = i + 1; j < n; j++) {
            if(arr[j] < arr[minIndex]) {
                minIndex = j;
            }
        }

        int temp = arr[i];
        arr[i] = arr[minIndex];
        arr[minIndex] = temp;
    }

    for(int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }

    return 0;
}
```

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

```
abhishekmishra28@Abhisheks-MacBook-Air Chapter Hello World % gcc selectionsort.c
abhishekmishra28@Abhisheks-MacBook-Air Chapter Hello World % ./a.out
5
70 20 90 10 40
10 20 40 70 90 %
abhishekmishra28@Abhisheks-MacBook-Air Chapter Hello World %
```